

Grievance Redressal Survey Results

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This project uses synthetic data inspired from the Grievance Redress Report (GRR-1) by Mott MacDonald Limited. The GRR report is the first output from the spot check and beneficiary feedback exercise of the National Cash Transfer Programme (NCTP) also known as the Benazir Income Support Programme (BISP).

Data Source= [Grievance Redress Report \(GRR-1\)](#)

The following flowchart illustrates the logical structure and branching pathways of the survey questionnaire.

 Questionnaire

```
In [13]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
plt.style.use('ggplot')
from texttable import Texttable
import plotly.io as pio
pio.renderers.default= 'notebook_connected'
import plotly.express as px
import plotly.graph_objects as go
```

```
In [14]: # Load data with a seperate dataframe for seperate sheets

# Load sheet with responses e.g Q1:a, Q2b
df_raw_responses= pd.read_excel("Clean_Grievance_Sample.xlsx", sheet_name='Responses')

#Load sheet with keys to map responses to labels e.g a --> Yes, b ---> No
df_raw_keys= pd.read_excel("Clean_Grievance_Sample.xlsx", sheet_name="Keys")
df_raw_keys= df_raw_keys.set_index("Options")

#Load sheet with question labels mapping to numbers e.g.
df_raw_questions= pd.read_excel("Clean_Grievance_Sample.xlsx", sheet_name="Questions")
```

```
In [15]: #Responses
df_raw_responses.head()
```

```
Out[15]:
```

	PROVINCE	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
0	AJK/GB	b	a	a	a	a	NaN	NaN	NaN
1	KP	b	a	b	NaN	NaN	b	a	NaN
2	KP	a	a	b	NaN	NaN	b	b	NaN
3	KP	c	b	b	NaN	NaN	b	c	NaN
4	Balochistan	b	a	a	a	a	NaN	NaN	NaN

```
In [16]: # Answer Keys
df_raw_keys
```

```
Out[16]:
```

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
--	----	----	----	----	----	----	----	----

Options	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Options								
a	Quality of Service Issues	In Person	Yes	Yes	Yes	Yes	Lack of time	Complaint resolved and I am satisfied
b	Payment Issues	Phone	No	No	No	No	Language barrier	Complaint resolved but I am not satisfied
c	Documentation Issues	Other	NaN	NaN	NaN	NaN	Didn't feel it would make a difference	Complaint not resolved
d	Eligibility Issues	NaN	NaN	NaN	NaN	NaN	Other	Other
e	Other	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
In [17]: # Questions
df_raw_questions
```

Number	Question
0	Q1 What was the reason behind your complaint?
1	Q2 How did you lodge a complaint?
2	Q3 Did anyone contact you?
3	Q4 Was your complaint resolved?
4	Q5 Are you satisfied with the resolution?
5	Q6 Did you complain again?
6	Q7 Why didn't you complain again?
7	Q8 What was the outcome of complaining again?

```
In [18]: # Create a nested dictionary mapping question numbers to answers

dict_map = {} # Create empty dictionary for question number: {answer:option label} pairs
for question_number in df_raw_keys.columns:
    dict_answers = {} # For each question, create an empty dictionary for the relevant answers
    df_answer_labels = df_raw_keys[question_number].dropna() # Drop missing values (when no answer)
    for option in df_answer_labels.index:
        dict_answers[option] = df_answer_labels[option]
    dict_map[question_number] = dict_answers

df = df_raw_responses.replace(dict_map) # Use the newly created dictionary to map relevant answers

new_colnames = list(df_raw_questions["Number"]+": "+df_raw_questions["Question"])
new_colnames.insert(0,"Province")
df.columns = new_colnames

df.head()
```

	Province	Q1: What was the reason behind your complaint?	Q2: How did you lodge a complaint?	Q3: Did anyone contact you?	Q4: Was your complaint resolved?	Q5: Are you satisfied with the resolution?	Q6: Did you complain again?	Q7: Why didn't you complain again?	Q8: What was the outcome of complaining again?
0	AJK/GB	Payment Issues	In Person	Yes	Yes	Yes	NaN	NaN	NaN

	Province	Q1: What was the reason behind your complaint?	Q2: How did you lodge a complaint?	Q3: Did anyone contact you?	Q4: Was your complaint resolved?	Q5: Are you satisfied with the resolution?	Q6: Did you complain again?	Q7: Why didn't you complain again?	Q8: What was the outcome of complaining again?
1	KP	Payment Issues	In Person	No	NaN	NaN	No	Lack of time	NaN
2	KP	Quality of Service Issues	In Person	No	NaN	NaN	No	Language barrier	NaN
3	KP	Documentation Issues	Phone	No	NaN	NaN	No	Didn't feel it would make a difference	NaN
4	Balochistan	Payment Issues	In Person	Yes	Yes	Yes	NaN	NaN	NaN

In [19]:

```

### Plot a Bubble Map to identify provinces with the highest frequency of complaints

# Create a dataframe for lat_long data
lat_long= pd.DataFrame({
    'Province': ["Punjab", "Sindh", "KP", "Balochistan", "AJK/GB"],
    'Lat': [31.1704, 25.8943, 34.9526, 28.4907, 34.8641],
    'Long': [72.7097, 68.5247, 72.3311, 65.0958, 74.4232]
}).set_index("Province")

# Join Province Counts with Lat Long data
data_map = df["Province"].value_counts().to_frame()
data_map = data_map.join(lat_long)
data_map = data_map.reset_index()
data_map.rename(columns={'index': 'Region'}, inplace=True)

# Plot Bubble map
fig = go.Figure()
fig.add_trace(go.Scattergeo(
    lat=data_map["Lat"], lon=data_map["Long"],
    marker = dict(
        size = data_map["Province"]*12,
        color='salmon',
        line_color='rgb(40,40,40)',
        line_width=0.7),
    showlegend=False
))

fig.add_trace(go.Scattergeo(
    lat=data_map["Lat"], lon=data_map["Long"],
    text=data_map["Region"],
    textposition="middle center",
    textfont=dict(size =13, color='black', family='serif'),
    mode='text',
    showlegend=False
))

fig.update_layout(
    title_text='Hotspots- Where Complaints are Most Frequent',
    titlefont=dict(size =30, color='black', family='serif'),
    title_x=0.5,
    geo=dict(scope='asia', lataxis_range=[23.9,37.5], lonaxis_range=[60.9, 78],
    landcolor = 'rgba(217, 217, 217,0.5)',
    showcountries = True,
    showcoastlines=True,

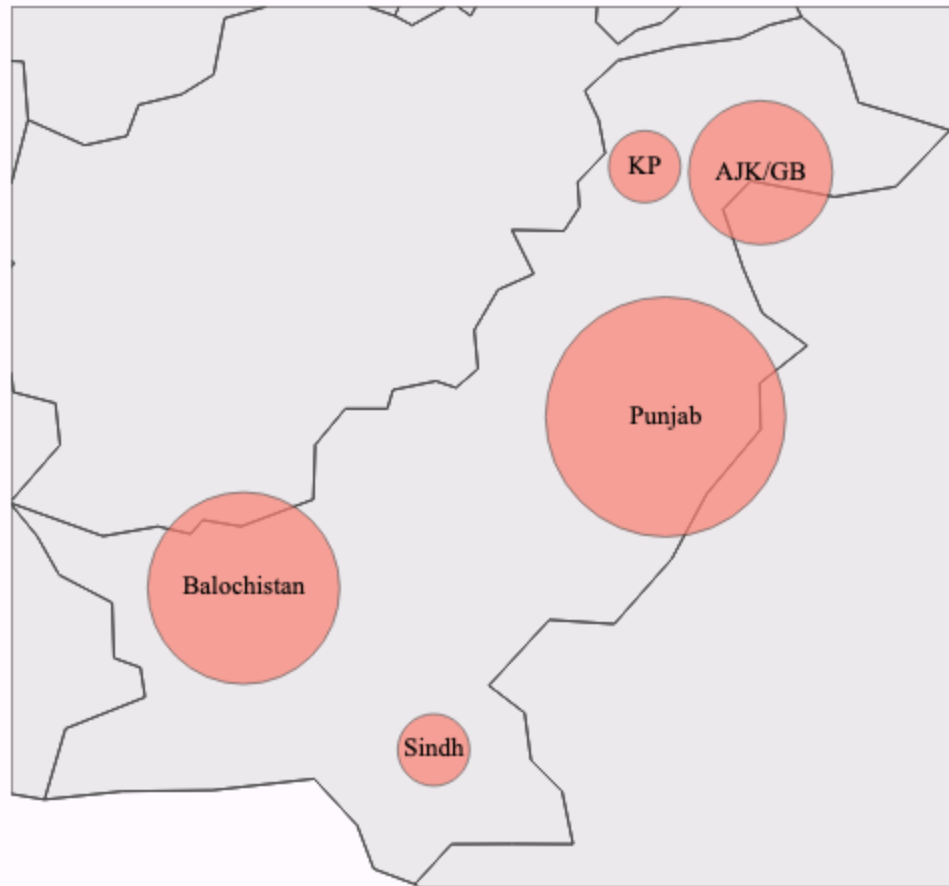
```

```

    bgcolor='rgb(255,250,255)'),
    paper_bgcolor='rgb(255,250,255)',
    margin={"r":10,"t":50,"l":10,"b":10}
)
fig.show("png")

```

Hotspots- Where Complaints are Most Frequent



The data indicates that the highest volume of complaints originated from Punjab and Balochistan. While the high frequency in Punjab correlates with its large population, the significant number of grievances from Balochistan is a notable outlier, given that it is the least populous province.

In [20]:

```

# Define a function to plot pie charts
def plot_pie(data,color_list,title,ax_num):
    """
    Generates and plots a pie chart based on the provided dataset.
    Args:
        df (pd.DataFrame): The dataframe containing the data to plot.
        colors (list[str]): A list of hex codes or color names for the pie slices.
        title (str): The title to display above the chart.
        ax (matplotlib.axes.Axes): The specific subplot axis where the chart should be drawn.
    Returns:
        matplotlib.axes.Axes: The axis object containing the plotted pie chart.
    """
    data.value_counts().plot(
        kind='pie',
        ax=ax[ax_num],
        colors= color_list,
        autopct='%1.1f%%',
        textprops={'fontsize': 12,'color':'darkblue','fontweight':550,'family':'serif'},
        wedgeprops = {'width':0.6,'linewidth':3, 'edgecolor':(1,1,1,1)})

    ax[ax_num].set_title(title,

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```

        loc='center', fontsize=25, fontweight="bold", family='serif')
    ax[ax_num].set_ylabel("")
    ax[ax_num].tick_params(axis='x', labelsizes=25)
    ax[ax_num].tick_params(axis='y', labelsizes=25)

```

In [21]:

```

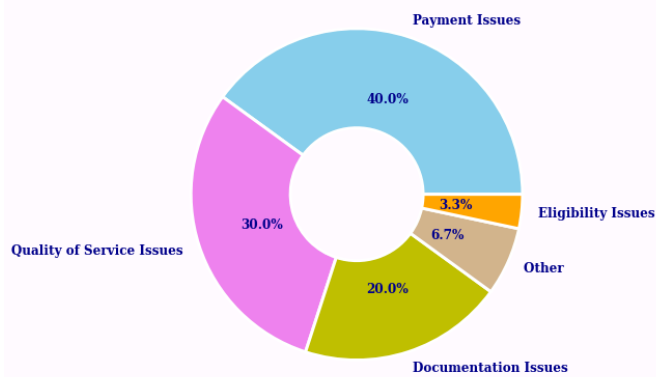
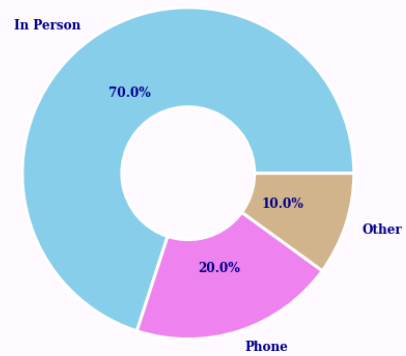
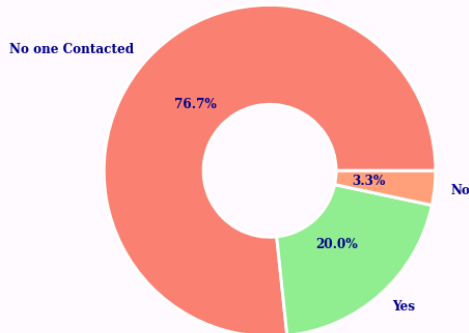
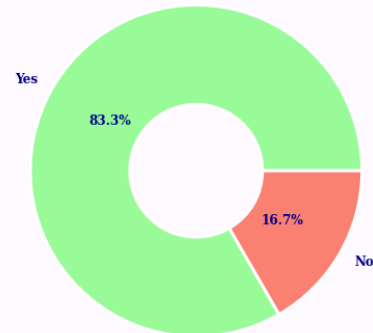
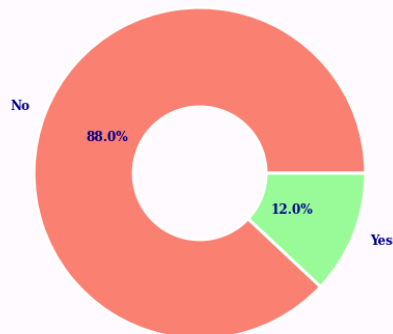
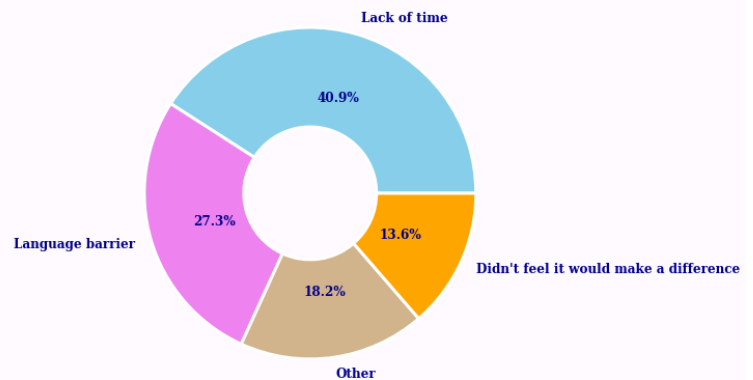
# Plot piecharts for Questions 1,2,4,5,6 and 7 using the function defined above
fig,ax= plt.subplots(3,2,figsize=(20,25))

ax = ax.flatten()

plot_pie(df["Q1: What was the reason behind your complaint?"],['skyblue', 'violet','y','tan'])
plot_pie(df["Q2: How did you lodge a complaint?"],['skyblue','violet','tan'],'Complaint Se')
plot_pie(df["Q4: Was your complaint resolved?"].fillna("No one Contacted"),['salmon', 'lightcoral'])
plot_pie(df["Q5: Are you satisfied with the resolution?"].dropna(),['palegreen','salmon'])
plot_pie(df["Q6: Did you complain again?"].dropna(),['salmon', 'palegreen'],'Did you complain again?')
plot_pie(df["Q7: Why didn't you complain again?"].dropna(),['skyblue', 'violet','tan','orange'])

fig.set_facecolor(tuple(np.array([255, 250, 255]) / 255))
plt.show()

```

Complaint Reasons**Complaint Source****Complaint Resolved?****Satisfied with Resolution?****Did you complain again?****Reasons for Not Complaining Again**

- **Primary Grievances:** Payment issues were the most frequent cause for complaints, followed by service quality and documentation requirements.
- **Lodgment Method:** The majority of complaints were submitted in person.
- **Responsiveness & Resolution:** Most complainants received no feedback from authorities. However, among the cases that were resolved, users reported high levels of satisfaction with the outcome.
- **Follow-up Barriers:** There was a low rate of repeat complaints, primarily due to time constraints and language barriers.

In [22]:

```
# Define function for plotting barchars
def plot_bars(col,barcolors):
    """
    Generates a bar chart for a specified data column.
    Args:
        column (pd.Series): The data column to be visualized.
        colors (list[str]): A list of color hex codes or names for the bars.
```

```

Returns:
    matplotlib.axes.Axes: The axes object containing the bar chart.
"""
data_bar= pd.crosstab(df[col],df["Province"],dropna=False)
data_bar["Total"] = data_bar.sum(axis=1)

#Convert numbers to percentages
data_bar= data_bar.apply(lambda r: r*100/r.sum(),axis=0)
data_bar= data_bar.round(decimals=1)

#Reorder columns
data_bar= data_bar.loc[:,["Punjab","Sindh","KP","Balochistan","AJK/GB","Total"]]

#Plot barchart
fig,ax= plt.subplots(figsize=(48,20))
data_bar.T.plot.bar(
    ax=ax, rot=0, legend=False,
    width=0.8,color=barcolors, edgecolor='k')

# Add this loop to add the annotations
for p in ax.patches:
    width = p.get_width()
    height = p.get_height()
    if height>0:
        x, y = p.get_xy()
        ax.annotate(
            height, (x + width/2, height+1.02),ha='center',
            fontsize=25,fontweight="bold", color='darkblue')

ax.legend(loc='best',prop={'size': 30})
ax.set_title("\n"+data_bar.index.name+"\n", loc='center', fontsize=50, fontweight="bold")
ax.set_xlabel('Region',fontsize=40, fontweight="bold",family='serif')
ax.set_ylabel('Percentage',fontsize=40, fontweight="bold",family='serif')
ax.tick_params(axis='x', labelsiz=25)
ax.tick_params(axis='y', labelsiz=25)

fig.set_facecolor(tuple(np.array([255, 250, 255]) / 255))
fig.tight_layout()

```

In [23]:

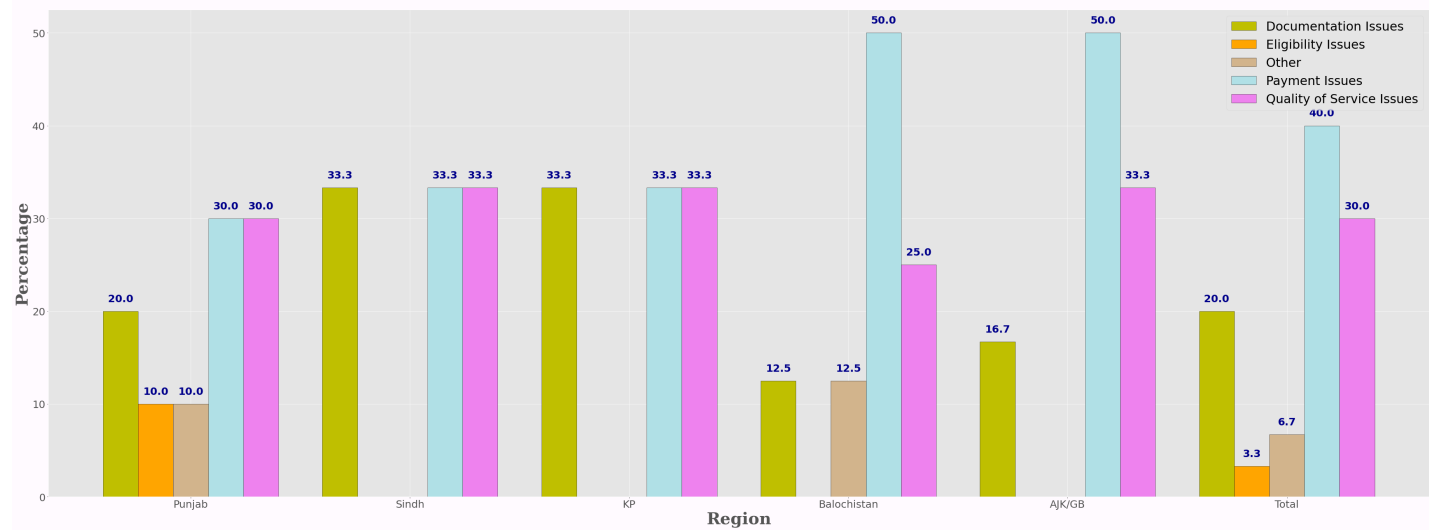
```

# Plot Bar charts

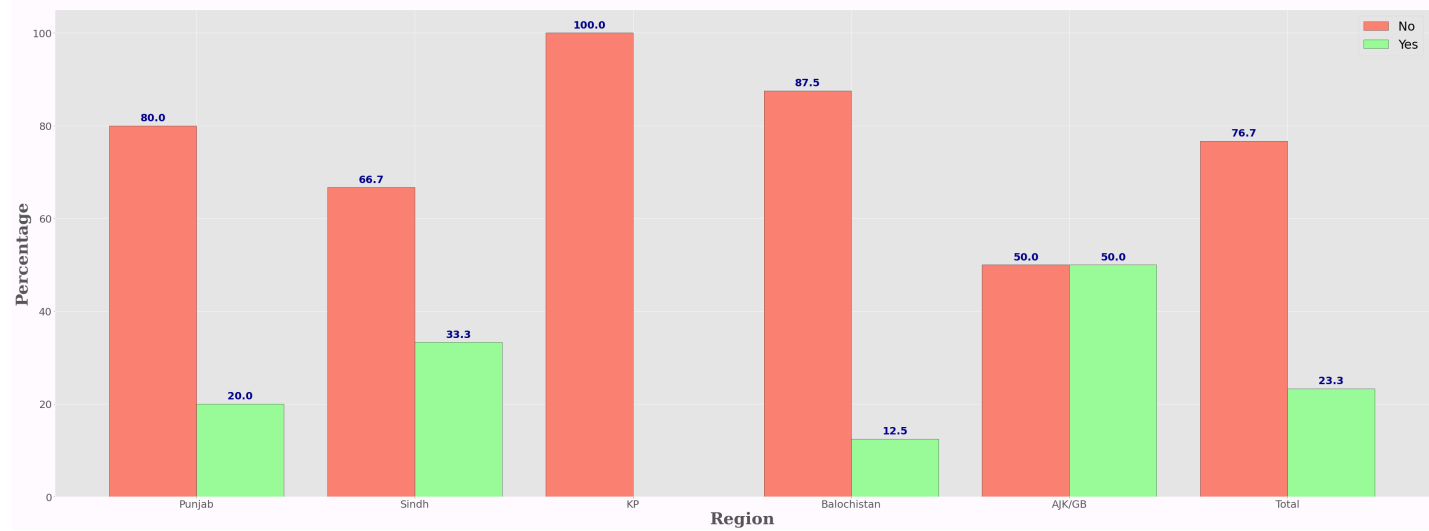
plot_bars(df.columns[1],['y','orange','tan','powderblue','violet'])
plot_bars(df.columns[3],{'Yes':'palegreen','No':'salmon'})
plot_bars(df.columns[4],{'Yes':'palegreen','No':'salmon'})
plot_bars(df.columns[7],['orange','skyblue','violet','tan'])

```

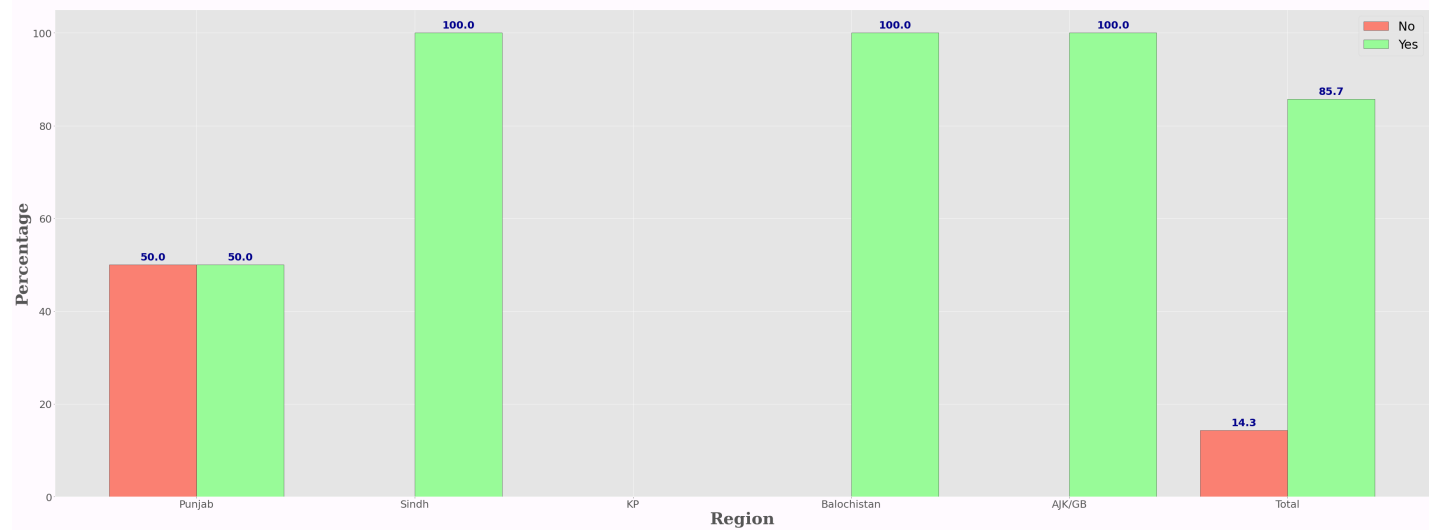
Q1: What was the reason behind your complaint?



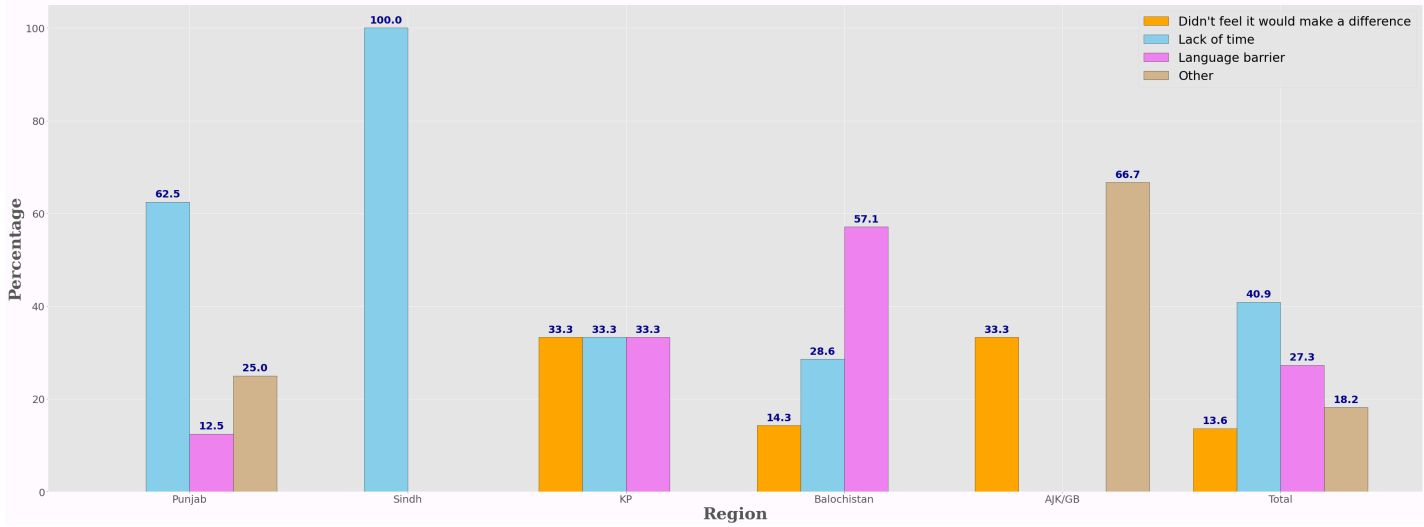
Q3: Did anyone contact you?



Q4: Was your complaint resolved?



Q7: Why didn't you complain again?



- **Dominant Issues:** Payment grievances were the most prevalent issue across most provinces. Specifically, in Balochistan and AJK/GB, they accounted for 50% of the total complaint volume.
- **Responsiveness Gap:** A critical lack of engagement was observed: 100% of complainants in KPK and over 80% in Punjab and Balochistan were never contacted. This highlights an urgent need for a performance audit and stricter oversight of the resolution staff in these regions.
- **Resolution Rates:** While most provinces achieved a 100% resolution rate for the complaints they handled, Punjab showed a significant shortfall, with 50% of cases remaining unresolved.
- **Accessibility Barriers:** In Balochistan, language was identified as the primary barrier to lodging complaints. Implementing translation services could be a key strategic intervention for authorities to improve service accessibility.

```
In [ ]: ! jupyter nbconvert --to webpdf Grievance_redressal_sampledata.ipynb
```